1. Identity

Document	STAR-PKG-STAR-TIMECODE-EN · v1.3 · 2026-10-16
Package	star-timecode
Version	0.1.0
Capability	star.time.ccsds-ascii-timecode
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_timecode

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_timecode-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_timecode-0.1.0-py3-none-any.whl	35d96f1e37114b29112bb65054a1460d4a1e6c4879e2db8254a295c6c4793d49
star_timecode.py	f19c8b3847601f8c18582d864cfdf47af255186e70e0ebd52533146350271c50
star_timecode-0.1.0-source.zip	6e58da74b60815ce31f4c6721c993687e85b6579f864ef9a06693384ec3a7947

5. What it does, and its limits

Strict reader and writer of the CCSDS 301.0-B-4 ASCII time codes A (calendar) and B (day of year), with the fraction kept as its digits; fields equal to CPython datetime and numpy.datetime64 on 600 instants and to astropy on the 331 between 1900 and 2100; refuses all 40 malformed or out-of-range texts of the corpus, of which the ISO parsers of those libraries accept 10 to 14.

Syntax and calendar only. Second 60 is accepted at 23:59 of any date: whether a leap second really occurred there is a question for a leap-second table (see `star_timescales` in this repository), not for this module. No time zones or offsets (the codes are UTC), no truncated forms (date only, no seconds), no conversion to seconds, Julian dates or other time scales, no years before 0001 or after 9999. The limit of 12 fraction digits is this module's, not the standard's.

- `parse(text)` accepts code A `YYYY-MM-DDThh:mm:ss[.d...][Z]` and code B `YYYY-DDDThh:mm:ss[.d...][Z]` and returns (year, month, day, hour, minute, second, fraction). The fraction is returned as its string of digits (up to 12): nothing passes through a float, and a code read and written again is the same text.
- `format_a` and `format_b` write the two codes from the same fields; `day_of_year` and `month_day` convert between the two date forms with the Gregorian leap-year rule, years 0001 to 9999.
- Measured on 600 instants written in both codes: calendar fields, microseconds and day of year equal to CPython `datetime` and `numpy.datetime64` on all 600 and to astropy on the 331 between 1900 and 2100. Of 40 malformed or out-of-range texts, all 40 are refused.
- Every function returns its result or raises `ValueError`: wrong shape (widths, separators, case, spaces, signs, offsets, non-ASCII digits), a date or time that does not exist, second 60 anywhere but 23:59, arguments of the wrong type.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_timecode_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-timecode}: CCSDS ASCII time codes A and B, strict},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_timecode/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

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